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Real-Time Density/Viscosity Sensing for Automated Drilling: Fluid Mixing in a Drilling Fluid Processing Plant

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Abstract

During drilling, monitoring of drilling fluid density and viscosity provides critical information that affects both the safety and efficiency of the drilling operation. Here, an in-tank real-time density/viscosity sensing platform was tested at a drilling fluid processing facility during mixing and reconditioning operations of non-aqueous fluids. Data was gathered continuously during six weeks of operation, capturing both fluid properties live inside a 300-barrel mixing unit. The tested system is an in-tank instrument comprised of four sensors deployed at different locations within the mixing unit. The sensors are electromechanical tuning fork resonators, able to obtain density and viscosity simultaneously. Each sensor was housed in a submersible enclosure along with a temperature sensor. Two sensors were suspended into the tank at the top and bottom of fluid level at the suction side of the tank, while the remaining two were located, in a similar configuration, at the center of the tank. The four sensors transmitted live data to the control room for live visualization by the operator. As deployed, the system collected data continuously for six weeks, during which several drilling fluid reconditioning operations were ran. Various fluids were tested during mixing, their densities ranging from 7 to 17 pounds-per-gallon (ppg). Reference density measurements were taken 3-4 times per day during each drilling fluid mixing period using a standard drilling fluid balance measurement. Funnel viscosity measurements were also taken. Full drilling fluid reports, including Fann 35 rheology, were produced once per testing day. The density was found to be, typically, within about 0.13-0.26 ppg of the drilling fluid balance measurements or less, providing very good quantitative drilling fluid weight information. However, the automated system provided real-time data, allowing the operators to visualize every step of the drilling fluid weighing process. The measured viscosity was found to be sensitive to the flowing conditions within the tank and, although it cannot be numerically compared to the funnel viscosity, it followed the expected trends during operation. The system showed potential for autonomous monitoring of viscosity and density during fast-changing operational conditions such as during mud mixing and for mud monitoring during drilling operations.

Introduction

Automation is at the core of most modern industrial processes. In most modern industries, advances in sensing and computing have brought about an explosion of data that can be harnessed through new data analytics tools, such as machine learning and others, in order to optimize and enhance processes. In the oil and gas industry, drilling still represents one of the highest risk activities both in terms of safety of rig personnel as well as the financial risks brought about from drilling problems arising during operations. These problems give rise to Non-Productive Time (NPT), which includes the costs incurred when drilling must be stopped to address problems in the rig due to the drilling activities. For operating companies, this can translate into a large percentage of the total drilling costs. A large percentage of this NPT can be related to so-called hole problems, which includes events like well control issues, lost circulation, well integrity problems, stuck pipe, etc. A key part of the drilling operation is the drilling fluid, which by proper tuning of its weight, or density, helps avoid several of these problems by maintaining consistent pressure against the formation and preventing well integrity problems, reservoir fluids from entering the wellbore, or drilling fluid to be lost into the formation. The density of the drilling fluid is carefully tuned to avoid these operational problems. Drilling fluid properties are also key in efficiently removing cuttings and hole cleaning, which is accomplished by proper control of the drilling fluid viscosity. Thus, the quality of the drilling fluid, in one way or another, touches upon most of the hole problems that give rise to NPT. Technologies that help reduce these costs by proper drilling fluid management could then have a significant impact to the business.

Current viscosity/density measurements are done according to API standards [1,2]. Standard practices involve collecting samples from the mud circulation system. This may happen only a few times per day depending on the current stage of the operation. Once collected, the viscosity is measured by using a Marsh funnel and the density is measured with a calibrated weight balance. Detailed rheological measurements, including gel strength tests, may be performed using a Couette cylinder rotational rheometer, such as a Fann 35. These measurements are performed even less frequently. These practices bring about several issues: 1) measurements are not performed frequently enough, 2) quality of the measurements may be inconsistent as it depends on the skill level of the technician carrying out the measurements, and 3) these manual measurements cannot be easily integrated into modern rig systems for autonomous operations. These issues limit the use of traditional measurements in their ability to detect any evolving problem that might be reflected in the physical properties of the drilling fluid, such as a gas kick.

Deployed System

The system, referred to through this document as *Mud Fork*, is an in-tank system for real-time viscosity/density measurements of drilling fluids. A detailed description of the system can be found in prior publications [3,4]. Briefly, the sensing element consists of a vibrating tuning fork resonator. Vibrating viscosity/density sensing is a well-established technique and has been used in other fields such as logging while drilling [5,6]. Our device consists of a titanium vibrating cup with the fork's prongs machined on its surface. An internal piezoelectric element is used for actuation and sensing of the fork's vibrations (see Figure 1-a). The electromechanical system is used as the frequency-defining element of an oscillator circuit. Due to positive feedback from the circuit, the tuning fork system is made to oscillate at its main resonance mode. When oscillations are sufficiently large, the feedback circuit is opened, allowing the system to *ring down*. A ring-down curve is obtained approximately every second, from where the decay time and frequency can be obtained simultaneously. From these two parameters, the density and viscosity of the fluid can be obtained using the well-established theory of a cylinder interacting with a viscous fluid [7]. For a vibrating structure in a viscous fluid, when the density of the fluid changes, the effective mass of the vibrating object changes due to the extra *inertia* acquired as the fluid is displaced, thus inducing a change in vibrational frequency. On the other hand, As the structure vibrates in the fluid, it loses energy to the fluid through viscous dissipation, thus changing the rate at which oscillations decay through the ring-down and the exponential

decay constant. The fluid structure-interaction equations relating the fluid density (ρ) and viscosity (μ) to the ring-down decay constant (α) and the resonance frequency (f_0) in the fluid are [6,7]:

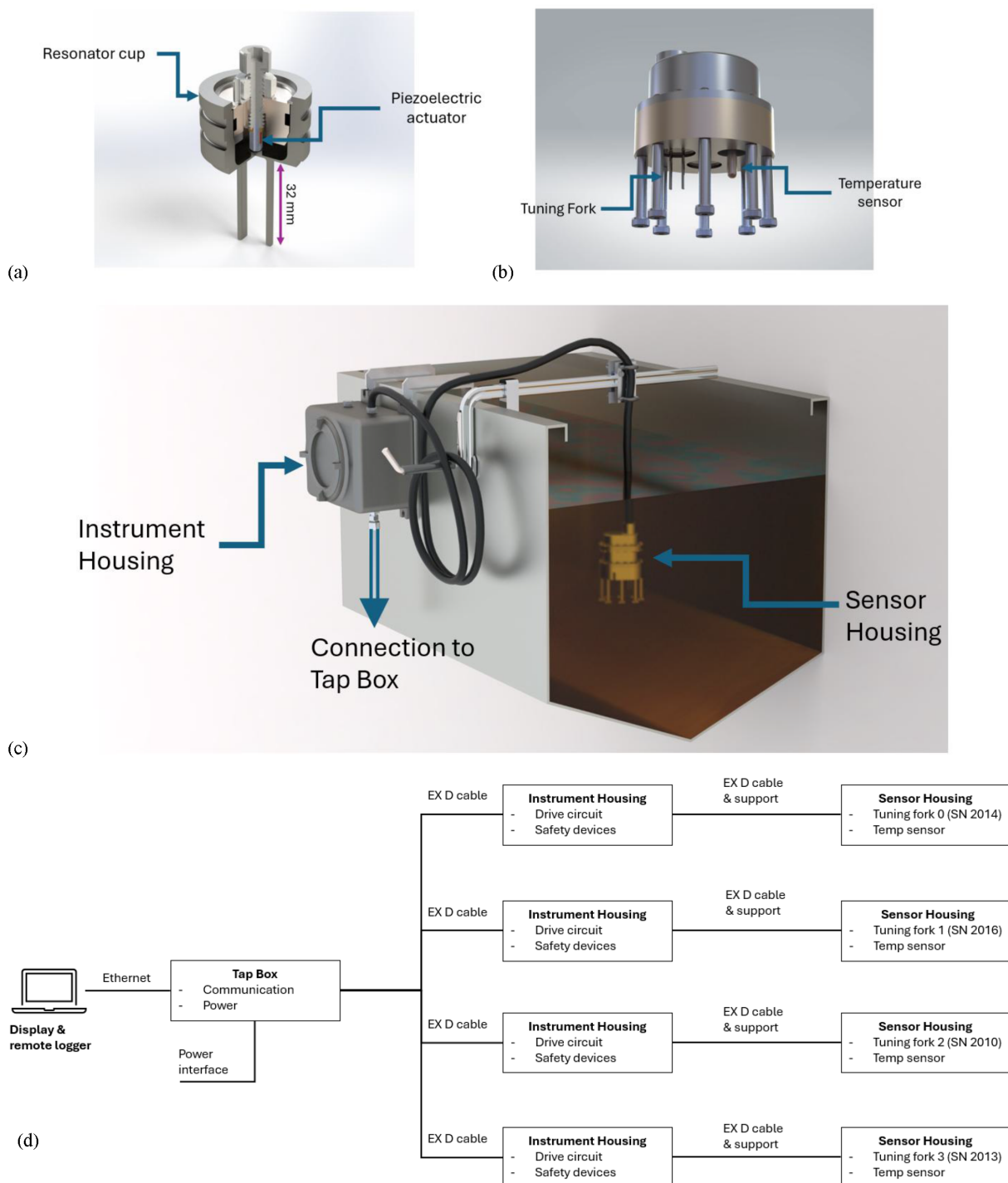


Figure 1—(a) Diagram of resonator cup. (b) Resonator cup in sensor housing. (c) Sketch of instrument and sensor housing deployed in a tank (d) System diagram for deployed Mud Fork system

$$\frac{1}{\alpha} = \frac{1}{2} \frac{CS}{m_{vac}} \left(\frac{f_o}{f_{vac}} \right)^2 \sqrt{\frac{\rho \mu f_o}{\pi}}$$

$$\left(\frac{f_{vac}}{f_o} \right)^2 = 1 + \frac{\beta}{\rho_{obj} \rho},$$

where ρ_{obj} is the density of the moving object, f_{vac} is the resonance frequency of the device in the absence of fluid, $m_{vac} = 0.25 \rho_{obj} V$ is the effective moving mass for the cantilevered motion of a single prong in vacuum, and S is the surface area of a single prong. The constants C and β are calibration constants that can be found empirically. We calibrate our sensors in mineral oil standard fluids for viscosity, with viscosity ranges between 1-100cP, and in calibrated samples of calcium bromide brines with concentrations corresponding to densities between 1-1.7gr/cc. Finally, the tuning fork resonator is installed in the sensor housing along with a commercial temperature sensor for in-tank deployment (see [Figure 1-b](#)).

Below, we focus on the overall system as implemented for deployment during the test at the mud processing facility.

The Mud Fork system consists of:

1. Sensor Housing containing the sensing element,
2. Instrument Housing driving the measurement
3. Tap Box providing power and telemetry to the whole system.

The sensing element of the instrument is an electromechanical tuning fork resonator. The sensor is deployed in a submersible housing inside the tank containing the fluid. The Sensor Housing is suspended in the tank by a ruggedized cable which connects the sensor to the Instrument Housing box, located outside of the tank. The Instrument Housing hosts the actuation/sensing electronics used for the measurements. The Instrument Housing connects to the Tap Box which takes raw data from the electronics box, calculates the viscosity and density, and sends the results to a display computer via wireless connection. The connection between the Instrument Housing and the Tap Box consists of DC power and an ethernet connection. The Tap Box contains a rugged Linux computer that does more advanced calculations, taking the raw timer measurements provided by the microcontroller and producing density and viscosity readings in specified units. The Tap Box also houses network equipment that facilitates the connections with displays and EDGE computing resources that are located off the rig structure. Finally, the TAP Box provides power to the entire system. Multiple Instrument Housing/Sensor Housing components can be connected to the Tap Box to obtain measurements from multiple tanks. [Figure 1](#) shows the configuration for the 4-Fork system developed as well as the serial number (SN) of each deployed sensor.

System Field Test Parameters

The chosen facility was a mud processing part in Madisonville, Texas. This site is operated by Newpark Fluid Systems. At the site, oil-based muds (OBMs), used at nearby wells in the Texas area, are brought for re-conditioning. Only OBMs are processed at this location. The location features a 300 bbl mud mixing unit where all mud reconditioning operations are carried out, as shown in [Figure 2](#). Storage tanks are used to keep the mud to be processed. When a reconditioning operation is to be executed, the mud from the storage tanks is moved into the mixing unit. A mixing hopper on the side is used to put any additives or weighing material in the mud. Inside the mixing unit, at the bottom of the tank, a series of jets are used to mix the fluid (see [Figure 3](#)). When mixing, fluid is pumped out of the tank through a flow line on the opposite side of the mixing hopper. The fluid is then re-circulated into the tank and injected into the mixing unit through the jet nozzles. The agitation of the fluid is done by the turbulent flows from the jets. Testing conditions are summarized in [Table 1](#).



Figure 2—(a-b) 300 bbl mud mixing unit where experiments were performed. (c) Mixing hopper from where samples were extracted. (d) Mud storage units from where mud is obtained and pumped into the mixing unit.

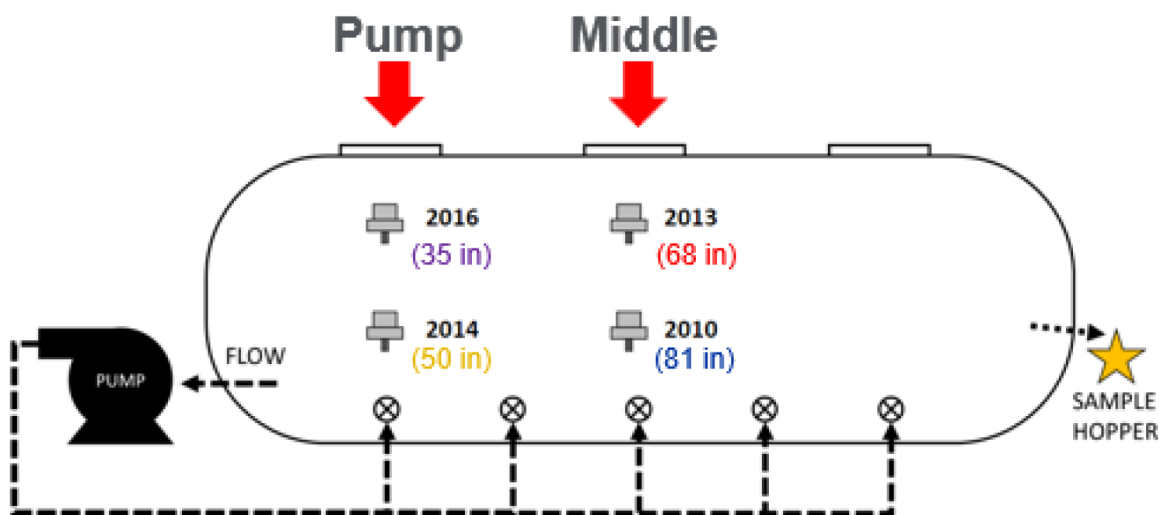


Figure 3—Deployment locations of the four sensor systems within the mud mixing unit. Note that below the serial number identifying each sensor, the distance from the top of the deck is shown. The relative distances within the diagram are not to scale.

Table 1—Testing conditions at mud processing facility

Mixing tank capacity	330 bbl
Fluids	OBM
Densities	55.4 – 128.7 PCF
Viscosities	6 – 103 cP (Fann 35 at 300rpm)
Flow rate / circulation speed	Unknown
Temperature	Ambient

A total of 4 sensor systems were deployed in mud mixing unit as shown in Figure 3. The four sensing units were suspended in the tank at different heights. Two sensors were suspended from the tank access port in the middle of the tank, labelled as Middle in Figure 3, while the other two sensors were suspended from the access port closed to the pump flow line, labelled as Pump. The distances from the deck at the top of the tank to the corresponding sensor are also shown in Figure 3 in parenthesis below each sensor label. The location of all sensors within the mixing tank are summarized in Table 2. The sensors were connected to their corresponding instrumentation housing which was located on the deck at the top of the mixing tank (see Figure 4), and all instrumentation housings were connected to the tap box, which was located at the foot of the tank as shown in Figure 4.

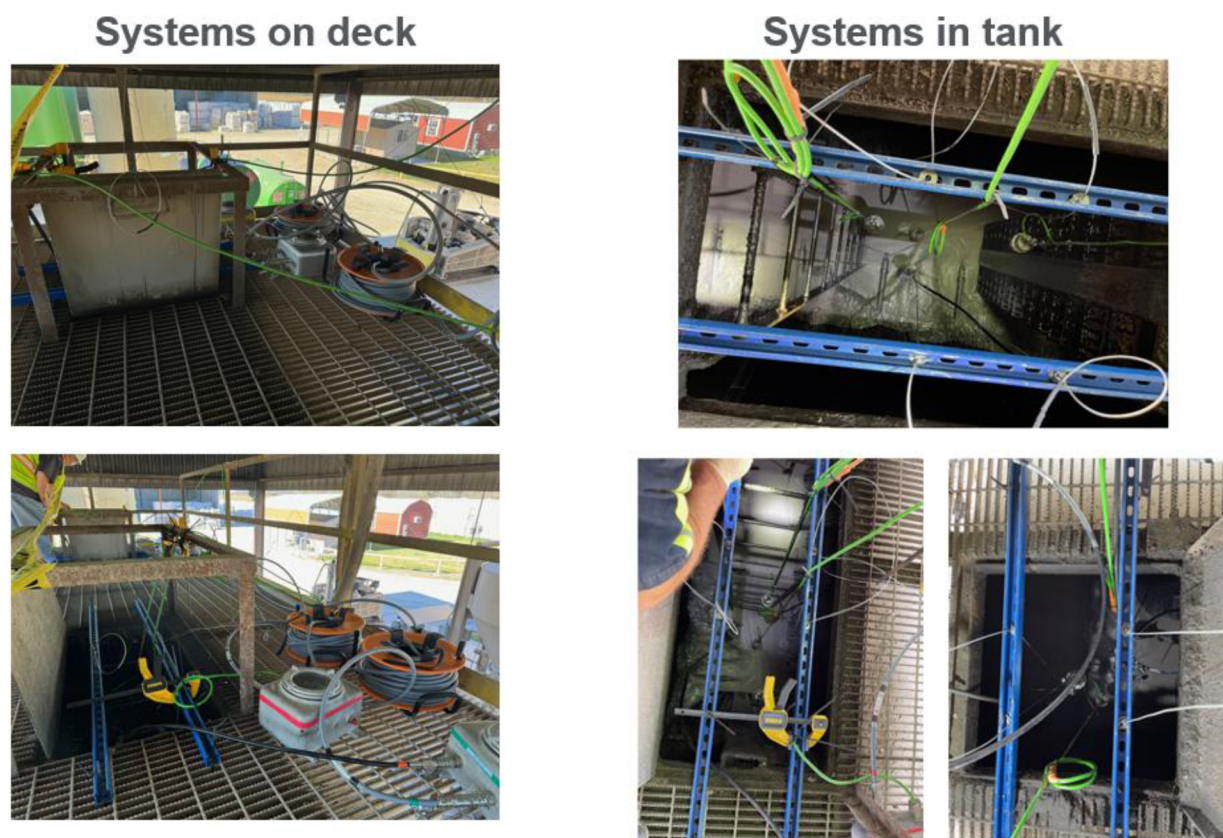


Figure 4—Systems deployed in mixing unit at mud processing facility

Table 2—Sensors deployed and their corresponding plot color code and depth. The table is organized as function of depth starting from the sensor at the largest depth.

Sensor	Plot Color	Depth (inches)
FORK2 / 2010	Blue	81
FORK3 / 2013	Orange	68
FORK0 / 2014	Yellow	50
FORK1 / 2016	Purple	35

System Field Test: Overview

After installation the system operated for a total of 44 continuous days. Because mud processing activities are not continuous at this facility (they only mix fluids when requested by a client), out of the 44 days in operation, 24 were days during which mud was mixed. After mixing operations, base fluid was kept in the tank where the sensors were deployed. Reference measurement tests occurred when mixing activities took place. During all operational days, all systems ran without any engineering intervention. No maintenance or cleaning was needed during the entire duration of the test. The system was left fully under the control of operators at the mud processing facility. The only incidents affecting operations were related to facility power outages due to bad weather.

During mud mixing activities, a dedicated mud engineer would take reference measurements. These measurements were taken by sampling from a sampling port near the mixing hopper. Reference measurements were taken mostly 3 times per day and a full mud report was generated once per day during test day. For the remaining daily measurements, a log of measurement results and/or daily events/activities was recorded.

System Field Test: Results

We show below some selected examples of the measured density and viscosity during a typical *mud mixing day*. Six days are shown here as representative examples of a typical day during mud mixing activities.

Typical Test Day Results

In this section we show data for a typical full day of testing. We show data for days 6, 7, 27, 28, 42 and 43. These are representative of the beginning, middle and final days of testing. Table 8 shows the sensors with their corresponding depth location and the color code used in the plots below.

Figures 5–8 show density and viscosity plots for days 5, 7, 27, and 43, correspondingly. Red diamonds on the density plots represent the reference measurements taken with the mud balance. During a typical operation mud was brought up to weight starting from a *base blend*. Thus, the main behavior seen through the test was that of a stepped density change as the fluid was brought up to the final composition. During this process, since the sensors were located at different heights, as the mixing tank was being filled and mixed with additives, different sensors came online in sequence, starting with the bottom-most sensor, Blue. Typically, after Blue, Orange, Yellow and Purple rapidly followed as they came in contact with the fluid. It should be noted though that Day 5 (Figure 5) shows a different order, with Blue and Yellow being the bottom-most sensors. This is due to a change in sensor heights that happened in this day. After this day, all sensor heights correspond to that in Table 2. For the viscosity plots in Figures 5–8, no reference is displayed. Currently, the viscosity measured by the Mud Fork system is relative, not absolute, since we cannot determine how much shear the fluid has experienced as it flows in the tank. However, a correlation between the Marsh Funnel viscosity and the Mud Fork was observed. With more data, likely a similar correlation may be possible with a Fann 35 rheometer in order to obtain an absolute viscosity measurement.

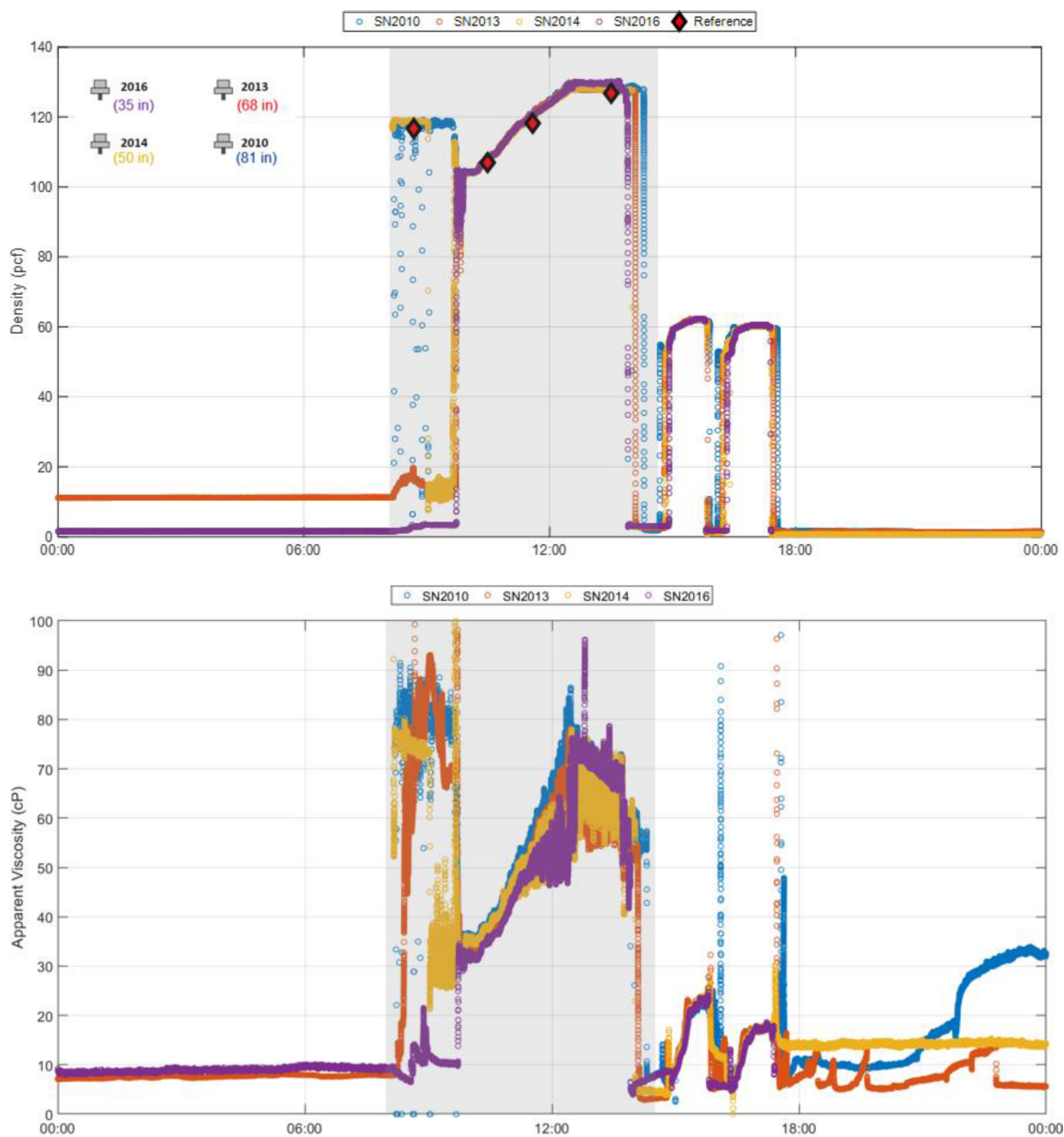


Figure 5—Day 5 of testing. The shaded gray area represents the timespan during which mud re-conditioning was carried out. During this time, mud weight was being tuned to a target mud density according to the facility's needs. In this day, mud started out with 15.6 ppg and cut it back to 13.7 ppg with diesel. Then, continued to weight up to 17 ppg. (Top) Density measurements in pound per cubic feet. The red diamonds represent the reference measurements taken using a standard mud balance. (Bottom) Viscosity measurements in centipoise. No reference measurements are shown as there is no direct calibration between Mud Fork viscosity and Marsh Funnel.

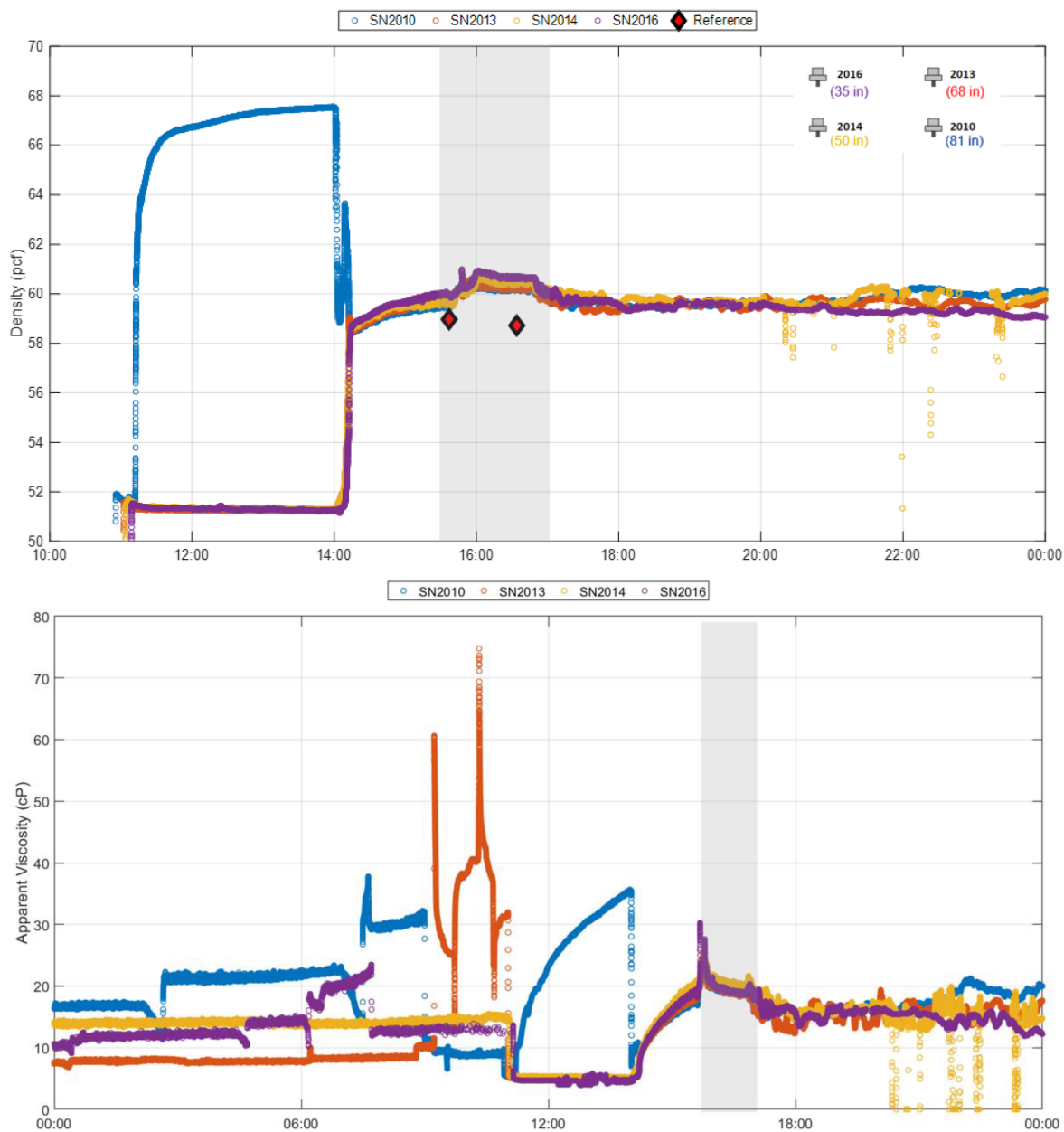


Figure 6—Day 7 of testing. The shaded gray area represents the timespan during which mud re-conditioning was carried out. During this time, mud weight was being tuned to a target mud density according to the facility's needs. Note that the time scale is different for the viscosity and density plots. Here, mixing of an unweighted blend started at approximately 14:00. (Top) Density measurements in pound per cubic feet. The red diamonds represent the reference measurements taken using a standard mud balance. (Bottom) Viscosity measurements in centipoise. No reference measurements are shown as there is no direct calibration between Mud Fork viscosity and Marsh Funnel.

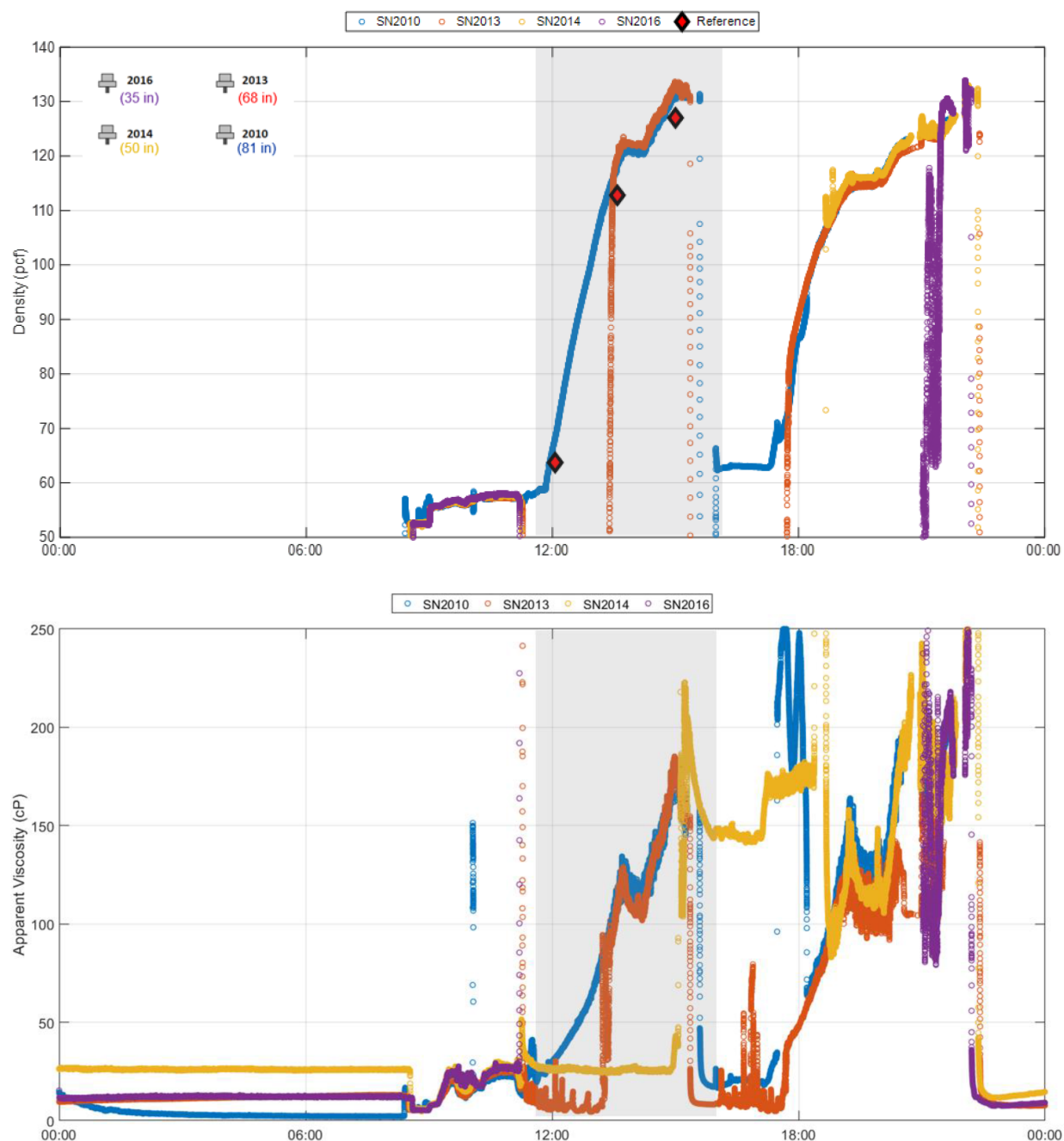


Figure 7—Day 27 of testing. The shaded gray area represents the timespan during which mud re-conditioning was carried out. During this time, mud weight was being tuned to a target mud density according to the facility's needs. Note that only sensors corresponding to blue and red traces are in the fluid level. Here, an 85/15 oil/water base blend was weighted up to 17ppg. (Top) Density measurements in pound per cubic feet. The red diamonds represent the reference measurements taken using a standard mud balance. (Bottom) Viscosity measurements in centipoise. No reference measurements are shown as there is no direct calibration between Mud Fork viscosity and Marsh Funnel.

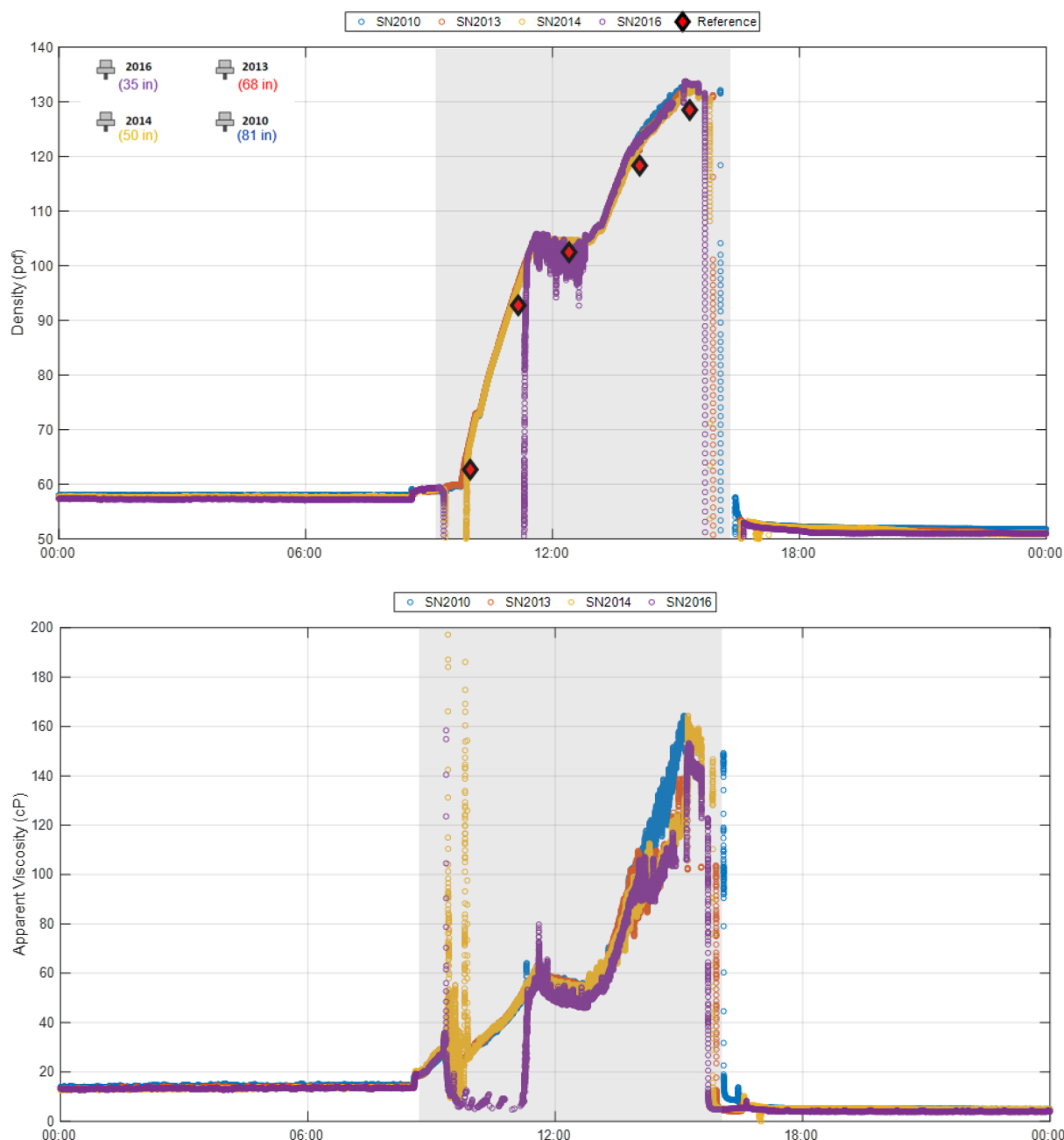


Figure 8—Day 43 of testing. The shaded gray area represents the timespan during which mud re-conditioning was carried out. During this time, mud weight was being tuned to a target mud density according to the facility's needs. In this day a 85/15 oil/water base blend that was left overnight in the mixing tank was used to weight up to 17.2ppg. (Top) Density measurements in pound per cubic feet. The red diamonds represent the reference measurements taken using a standard mud balance. (Bottom) Viscosity measurements in centipoise. No reference measurements are shown as there is no direct calibration between Mud Fork viscosity and Marsh Funnel.

Results Discussion

Figure 9 shows a plot of the density measurement difference with respect to the reference taken by the four Mud Fork systems close to a time when reference measurements were obtained. Measurements that were taken within a time window of ± 5 minutes from a reported sample collection time were included in the plot. The top of Figure 9 shows a lower band of measurements with a high density difference. These measurements correspond to times when the specific sensor was out of the fluid. During mixing, the tank level would sometimes drop, leaving the top-most sensors (or even all sensors) out of the fluid. The bottom of Figure 9 shows the data after removing the *out-of-fluid* measurements. The typical density

difference was typically less than 2 PCF (pounds per cubic feet), with a few additional measurements with density differences between 2-8 PCF. Figure 10 shows a parity plot between the measured and reference density measurements for all measurements within the ± 5 -minute time window when a reference sample was captured. Again, here, all *out-of-fluid* measurements were disregarded. The plot shows very good agreement between the measured and reference data, as shown by the proximity of the data to the ideal measurement with no error represented by the red dashed line.

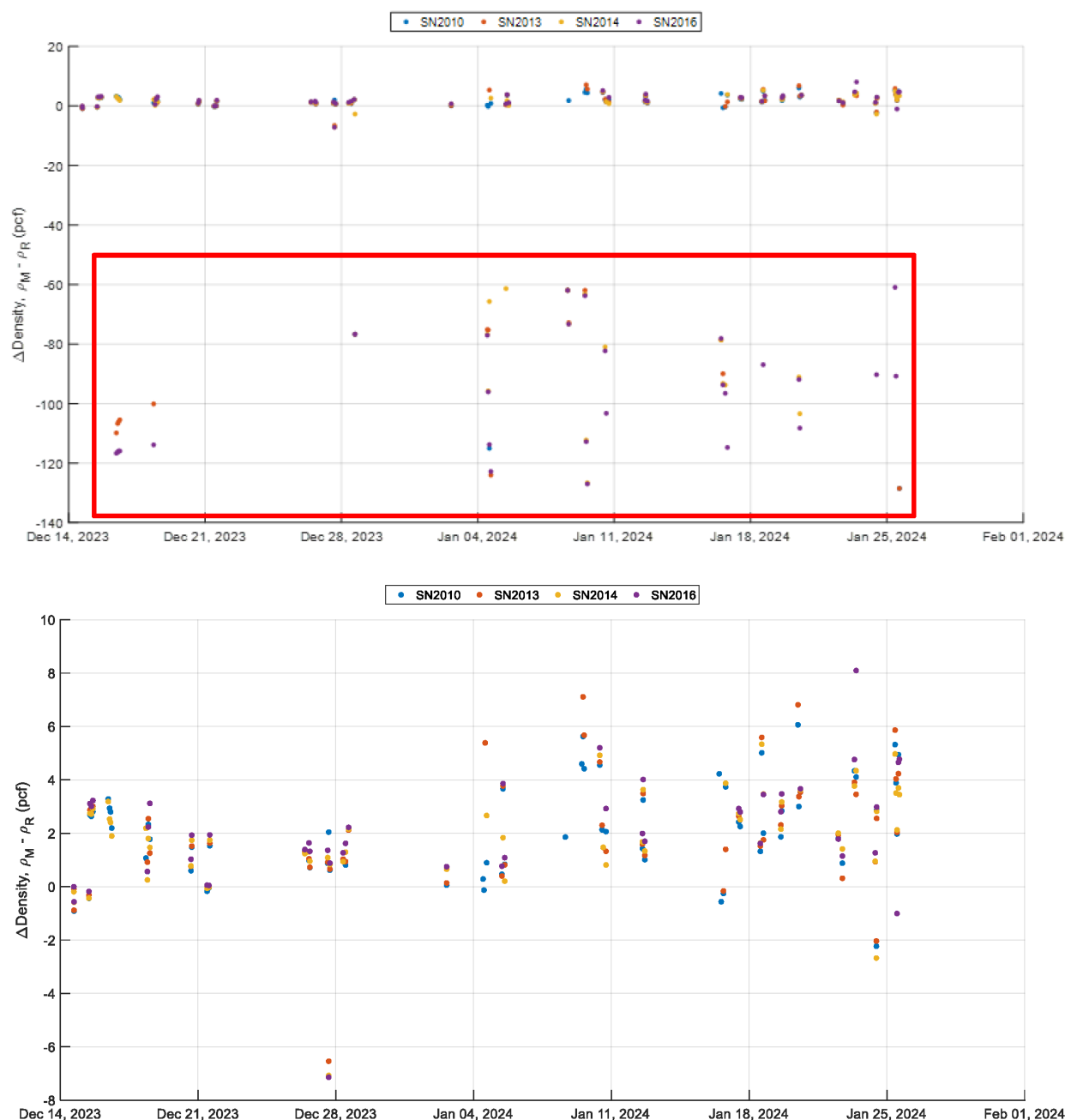


Figure 9—Measured density minus reference density over time where reference measurements were acquired. (Top) all data, including data points when sensors were not in the fluid. Red box encloses the points when the sensors were not in the fluid. (Bottom) remaining data after *out-of-fluid* points were removed.

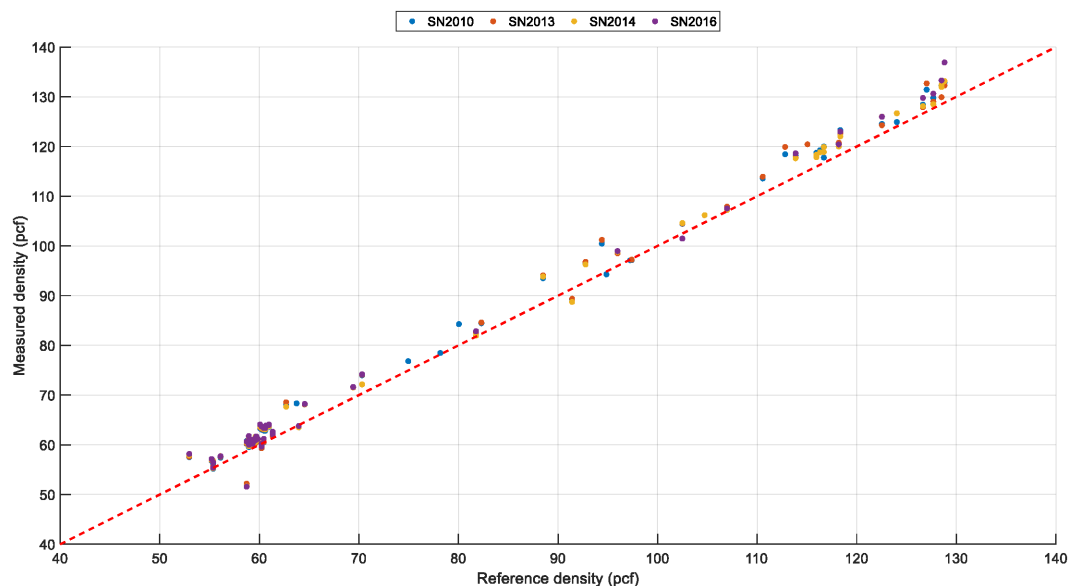


Figure 10—Measured density versus reference density for all measurements taken within a ± 5 -minute time window from a captured sample for density reference measurement. The red dashed line indicates a perfect correspondence between measured and reference density.

A direct comparison between measured and reference viscosity is more challenging due to the non-Newtonian, or shear-dependent viscosity, of drilling fluids. Figure 11 shows a plot of measured viscosity (in centipoise) versus Marsh funnel time (in seconds). It is interesting to note that despite not being directly comparable these two measurements still show a correlation between them, opening opportunities in the future for calibrating the tuning for measurement to the Marsh Funnel measurement. Despite not being as useful as a full rheological characterization typically provided by the Fann 35 field rheometer, this correlated value could still provide a relative measurement of viscosity useful as a quality control parameter. For absolute measurements of viscosity, it is recommended that in future tests the relation between flow past the tuning fork and measured viscosity is studied. This can be accomplished by tests in controlled flows at a flow loop facility.

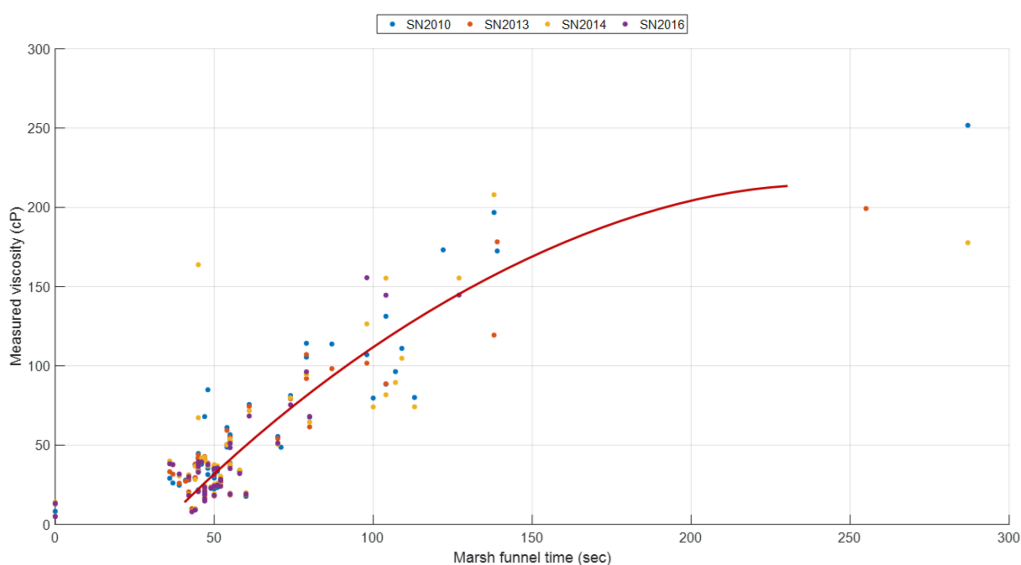


Figure 11—Measured viscosity versus Marsh Funnel time for all measurements taken within a ± 5 -minute time window from a captured sample for reference measurements. The red solid line is a guide to the eyes to highlight the correlation between the two values.

Conclusions

We presented here results from a major testing of the current performance of the automated density/viscosity system. The system was tested at a mud processing plant, where it operated continuously for 44 days, without any engineering support, demonstrating the systems capabilities for autonomous operations. Operators at the mud processing facility were able to use the system without any impact to their day-to-day operations. The performance of the system was qualified. Three main aspects of the system performance were studied: 1) overall system accuracy under realistic conditions, 2) system performance drifts that may be indicative of sensor fouling or need for cleaning, and 3) measurement dependence on sensor location in tank. In the first case, the sensor demonstrated excellent accuracy with a typical density measurement within 2 PCF from reference measurements. The system was not found to drift in performance during the 44 days of operation. No cleaning or maintenance was done to any of the systems during this time. No measurement dependence was found on tank location, although more studies could be done at a future time in larger tanks, where flows could be more inhomogeneous. For use during drilling operations, the tested system can be integrated into a rig SCADA system for real-time display of density and viscosity. This could dramatically increase the amount of data available to operators, enabling detection of anomalous mud properties that may be indicative of an undetected operational problem, such as gas kicks or loss of circulation events. An interesting prospect, due to the increased availability of data is to leverage machine learning algorithms to integrate these data with that of other available rig sensors in order to enhance early detection of these problems. Non-productive time (NPT) is a costly expense to operators. By reducing the number of unwanted events leading to NPT, operators can not only operate more safely, but also more efficiently.

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